

SIM800A GSM Module - Complete Study Guide

1. Introduction

SIM800A is a GSM/GPRS module used for communication over 2G networks. It can send SMS, make calls, and transmit data using AT commands.

2. Power Requirements

Voltage: 3.4V to 4.4V (ideal ~4V)

Idle Current: 10–30 mA

Transmission Peak Current: 1–2 A

Input Power during burst: 4–8 W

3. Working Overview

Battery supplies power → Module processes commands → Connects to GSM network → Transmits data via RF waves.

4. AT Commands Example

AT → Check communication

AT+CMGF=1 → Set SMS mode

AT+CMGS="+91XXXXXXXXXX" → Send SMS

Type message → Ctrl+Z

5. GSM Communication

Uses GSM standard with TDMA and GMSK modulation.

Frequency bands: 850/900 MHz

6. Burst Transmission

Transmission occurs in short bursts (~577 microseconds).

High current (2A) during burst, low current during idle.

7. RF Power

RF Output Power: ~2 W

Measured at antenna port with 50 ohm load.

2 W = 33 dBm

8. Signal Transmission

Electrical power → RF amplifier → Antenna → Electromagnetic waves.

Only small fraction reaches tower due to spreading loss.

9. Key Concepts

$P = V \times I$

RF power is not same as electrical input power.

System efficiency causes losses.

10. Practical Tips

Use proper power supply (2A capable).

Add capacitors (~1000uF).

Use proper antenna (50 ohm).